

Objective

- ❖ To design and fabricate a working electric motorcycle from a used petrol motorcycle.
- ❖ To study the performance parameters such as speed, torque, range, and charging time.
- ❖ To promote sustainable and eco-friendly transportation.
- ❖ To provide a low-cost, do-it-yourself (DIY) conversion method for local users.

Main Components

1. BLDC Motor
2. Li-ion Battery
3. Motor Controller
4. Throttle Assembly
5. Display Unit
6. Base Motorcycle Frame

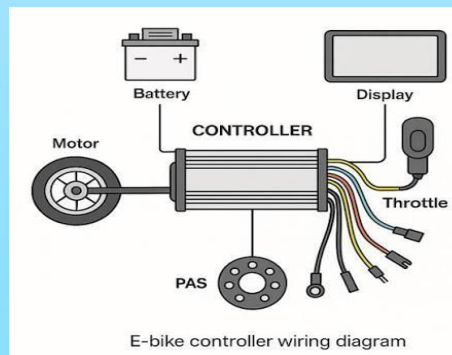


Introduction

- ❖ This project focuses on converting a conventional petrol-powered motorcycle into an electric motorcycle.
- ❖ The conversion replaces the internal combustion engine with an electric motor, battery pack, and controller.
- ❖ It reduces fuel dependence, lowers operating costs, and minimizes environmental pollution.
- ❖ The design is simple, cost-effective, and suitable for workshop-level fabrication

Working Principle

- ❖ The battery supplies power to the motor controller.
- ❖ The controller regulates power to the motor based on throttle position.
- ❖ The motor rotates and transmits torque through a reduction drive (sprocket/chain) to the rear wheel.
- ❖ The display shows real-time data such as speed and battery voltage.



**Technological University
(Mandalay)
Department of Mechanical
Engineering**

“Conventional Motorcycle Conversion to An Electric Motorbike”



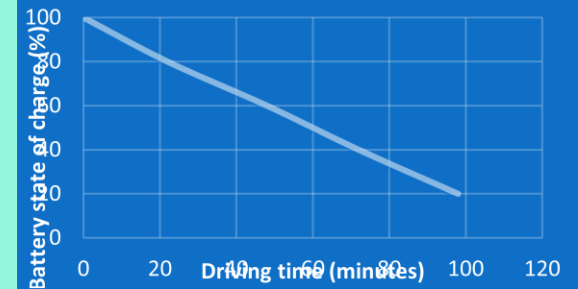
Challenges Faced

- ❖ Balancing battery weight without affecting handling.
- ❖ Matching motor RPM with wheel RPM through correct sprocket ratio.
- ❖ Ensuring proper cooling for motor and controller.
- ❖ Waterproofing electrical components for real-world riding.
- ❖ Sourcing affordable but reliable components locally.

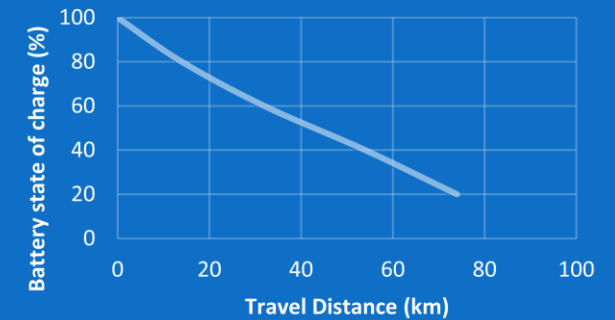
Technical Specification

1.	Motor Power	840.3	W
2.	Battery Voltage	64	V
3.	Battery Capacity	20	Ah
4.	Battery Energy	1280	Whr
5.	Estimate Time	1.52	hr
6.	Estimate Range	68.6	km
7.	Charging Time	4	hr
8.	Energy Consumption	18.66	Wh/km

Fabrication



Performance Testing



Advantages of Electric Bike

- ❖ Zero fuel cost – runs entirely on electricity.
- ❖ Very low noise and vibration.
- ❖ No engine oil, air filter, or spark plug changes – low maintenance.
- ❖ Environmentally friendly – zero tailpipe emissions.
- ❖ Can be charged from any standard household outlet.
- ❖ Ideal for short commutes, campus transport, and urban use.